



Data and Artificial Intelligence

Cyber Shujaa Program

Week 5 Assignment
Data Visualization Using Tableau - HR Analytics
Dashboard

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1.0 INTRODUCTION

This project involves developing an interactive HR analytics dashboard using Tableau. The aim is to provide actionable insights into employee demographics, performance, and compensation. This report outlines the development of both a summary and detailed employee view.

Objectives

The objectives of this project are to:

- Understand business needs
- Load and transform data
- Create calculated fields
- Design interactive visualizations
- Publish to Tableau Public (See downloaded project package for Tableau Desktop use)
- Document the process for a portfolio showcase

2.0 TASK COMPLETION

2.1 Understanding HR Business and Client Needs

Business Context:

The HR Manager requires a comprehensive dashboard that allows monitoring of workforce distribution, gender diversity, salary analysis, and individual employee records.

Client Requirements:

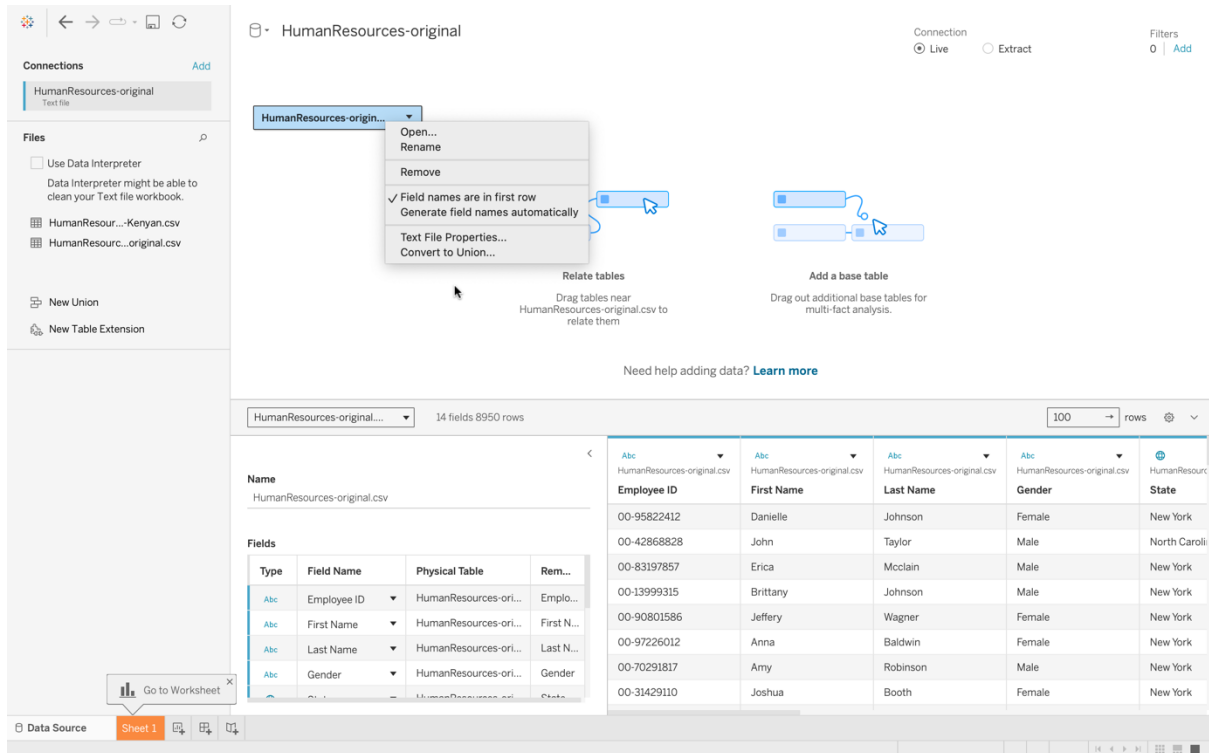
- Summary dashboard with Overview, Demographics, and Income Analysis
- Detailed list of employee records with filtering
- Interactive visuals (bar charts, maps, heat maps, etc.)
- Filtering by department, gender, education, etc.
- DSRN (Daily Sellable Room Nights)
- DBRN (Daily Booked Room Nights)
- DURN (Daily Utilized Room Nights)

2.2 Load Data

Steps Taken:

- Downloaded data from:
[Tableau HR Dataset](#)
- Extracted and opened files in Tableau
- Used Tableau's data interpreter to clean and understand relationships

Screenshot of loaded datasets in Tableau



2.3 Transform Data

Datatypes assessment was done and transformation of data made where needed.

Screenshots of data transformation

HumanResources-original.csv

Field separator: Semicolon
Text qualifier: Automatic
Character set: UTF-8
Locale: English (United Kingdom)

Relate tables
Drag tables near HumanResources-original.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

HumanResources-original... 14 fields 8950 rows

Employee ID	First Name	Last Name	Gender	State
00-95822412	Danielle	Johnson	Female	New York
00-42868828	John	Taylor	Male	North Caroli
00-83197857	Erica	Mcclain	Male	New York
00-13999315	Brittany	Johnson	Male	New York
00-90801586	Jeffery	Wagner	Female	New York
00-97226012	Anna	Baldwin	Female	New York
00-70291817	Amy	Robinson	Male	New York
00-31429110	Joshua	Booth	Female	New York

Go to Worksheet

Sheet 1

Department	Job Title	Education L.	Measure	
Customer Service	Customer Service Manager	Bachelor	Abc	
	Master	Abc		
	PhD	Abc		
	Customer Service Representative	High School	Abc	
	Help Desk Technician	Bachelor	Abc	
Finance	Support Specialist	High School	Abc	
	Accountant	Bachelor	Abc	
	Accounts Payable Speciali...	Bachelor	Abc	
	Finance Manager	Master	Abc	
	Financial Analyst	Bachelor	Abc	
HR	HR Assistant	Bachelor	Abc	
	HR Coordinator	High School	Abc	
	HR Manager	Master	Abc	
	Recruiter	Bachelor	Abc	
	IT Manager	High School	Abc	
IT	IT Support Specialist	PhD	Abc	
	Software Developer	Bachelor	Abc	
	System Administrator	Master	Abc	
	Marketing	Content Creator	Bachelor	Abc
		High School	Abc	

55 marks 55 rows by 1 column

2.4 Create Calculated Measures

Key Calculations:

- Age Groups:

```

IF [Age] < 25 THEN '<25'
ELSEIF [Age] >= 25 AND [Age] < 35 THEN '25-34'
ELSEIF [Age] >= 35 AND [Age] < 45 THEN '35-44'
ELSEIF [Age] >= 45 AND [Age] < 55 THEN '45-54'
ELSEIF [Age] >= 55 THEN '55+'
END

```

- **% Highlight Max:**

```
WINDOW_MAX([% Total Hired]) = [% Total Hired]
```

- **% Total Hired:**

```
[Total Hired] / TOTAL([Total Hired])
```

- **% Total Terminated:**

```
[Total Terminated] / TOTAL([Total Terminated])
```

- **Length of Hire:**

```

IF ISNULL([Termdate])
THEN DATEDIFF('year', [Hiredate], TODAY())
ELSE DATEDIFF('year', [Hiredate], [Termdate])
END

```

- **Status:**

```

IF ISNULL([Termdate]) THEN 'Hired'
ELSE 'Terminated'
END

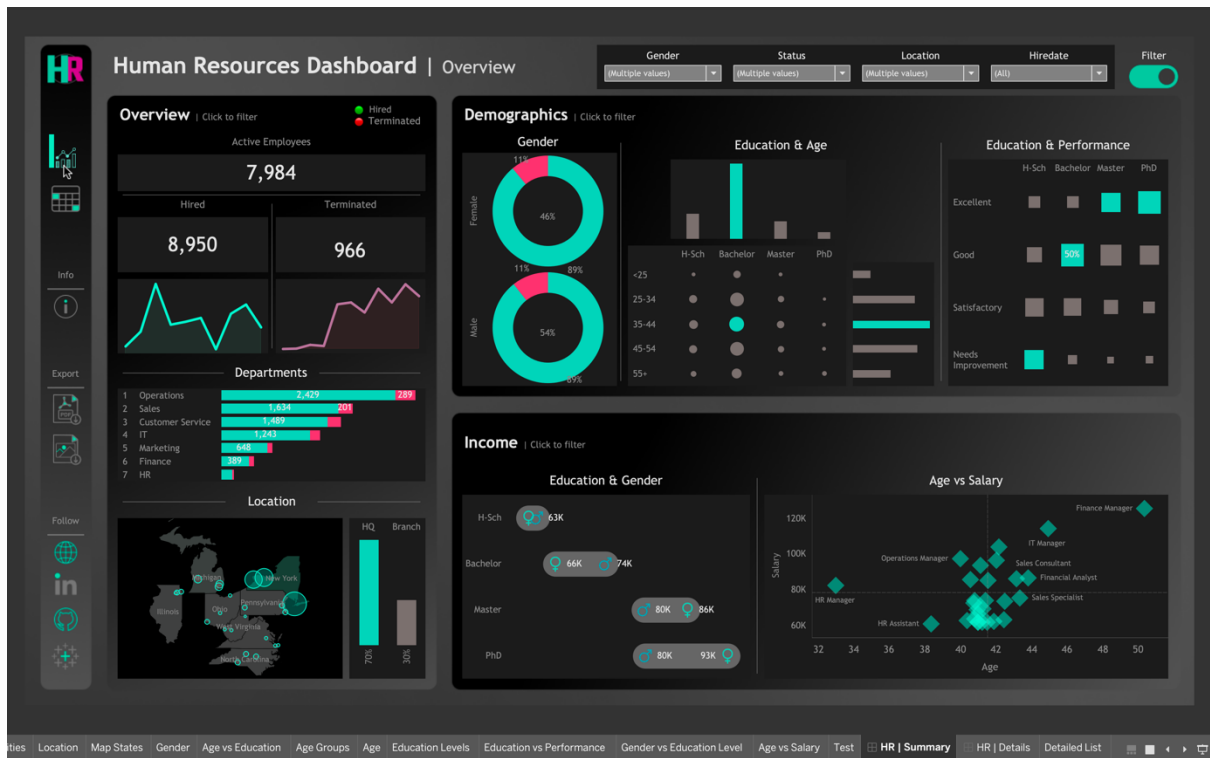
```

- **Salary:** Bins created for salary distribution

- Termdate, Gender, Avg Salary by Department, etc.

2.5 Visualization Dashboards

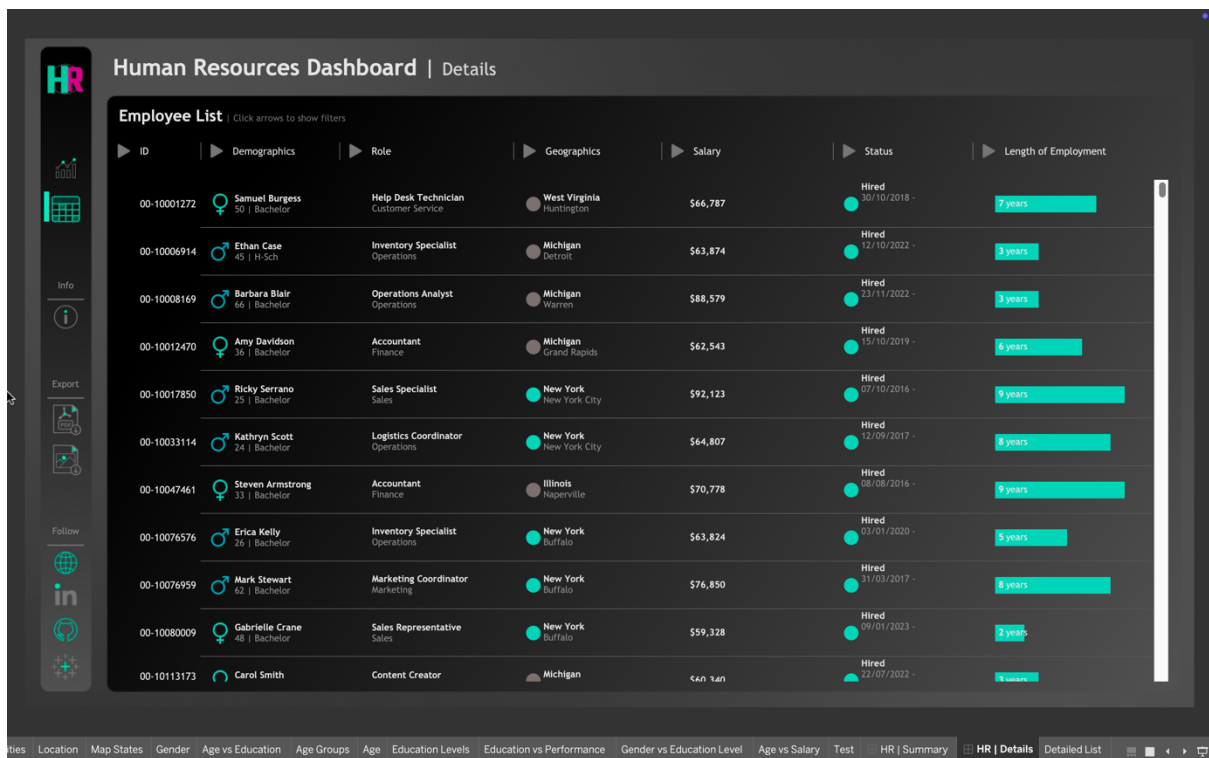
Summary Dashboard



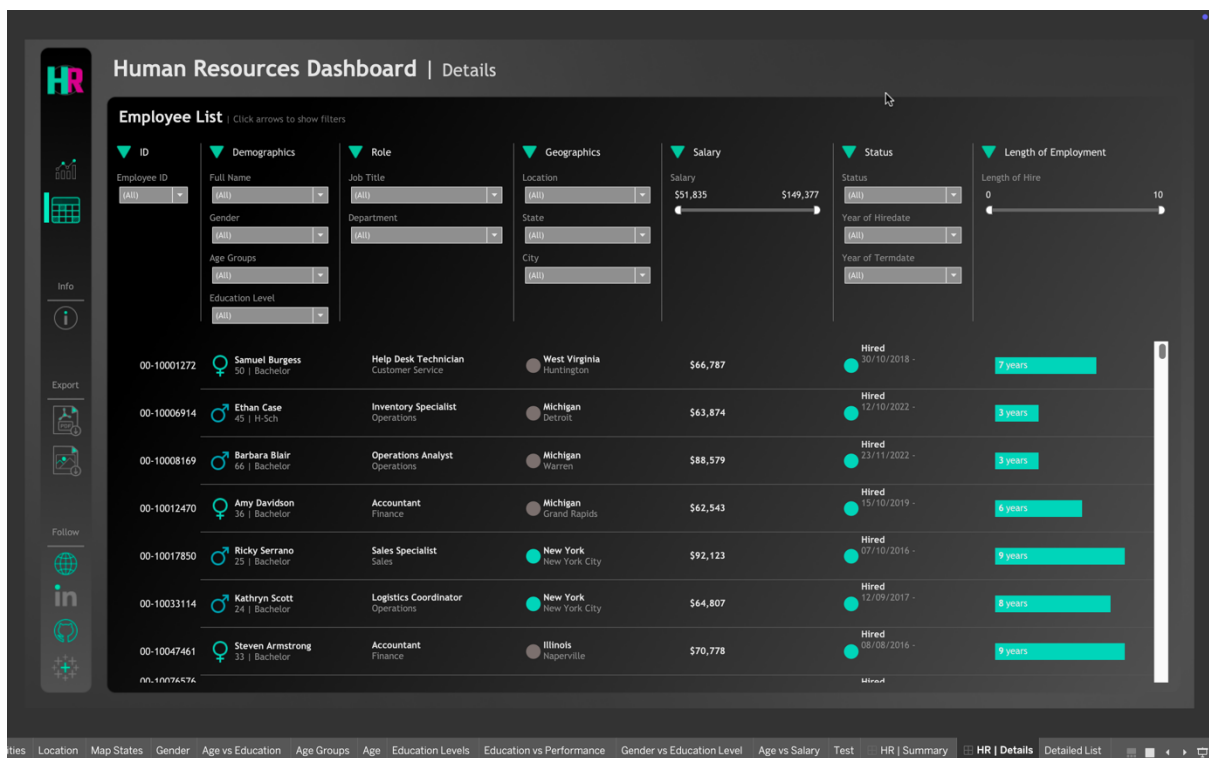
Summary Dashboard Overlay (Figma Gradient)



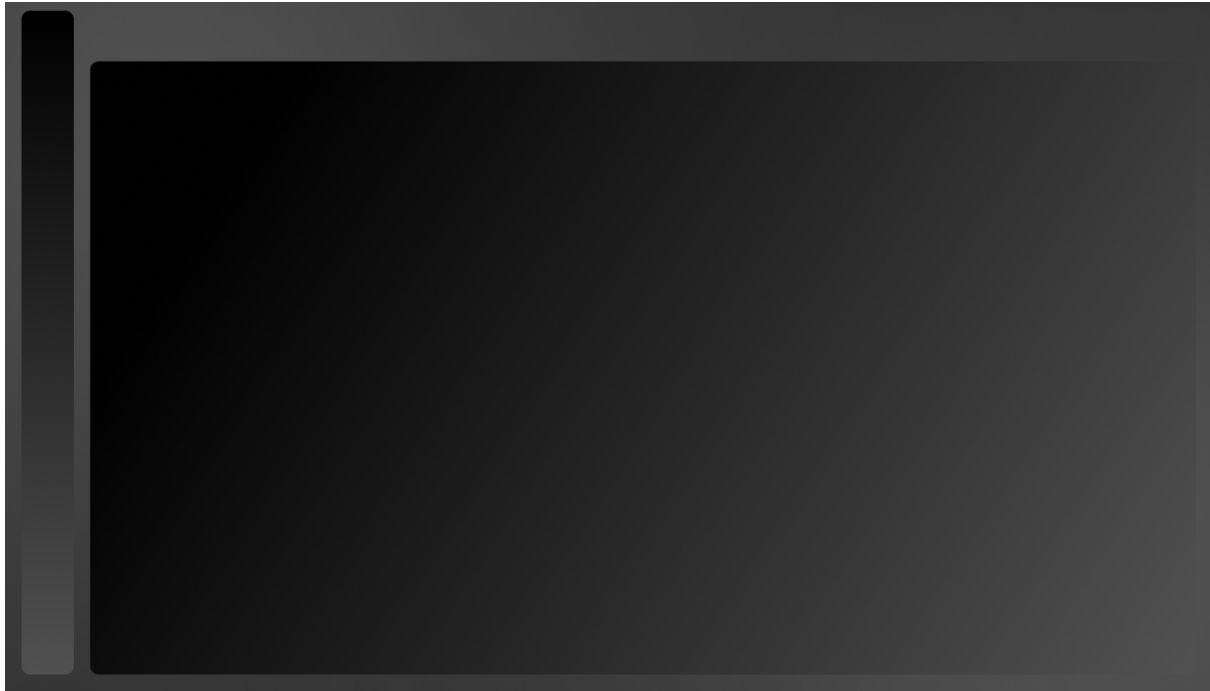
Detailed Dashboard (Filters invisible)



Detailed Dashboard (Filters visible)



Detailed Dashboard Overlay (Figma Gradient)



2.6 Publish and Share

To make the HR Analytics Dashboard accessible online, the Tableau package workbook (.twbx) was uploaded to Google Drive with public access enabled.

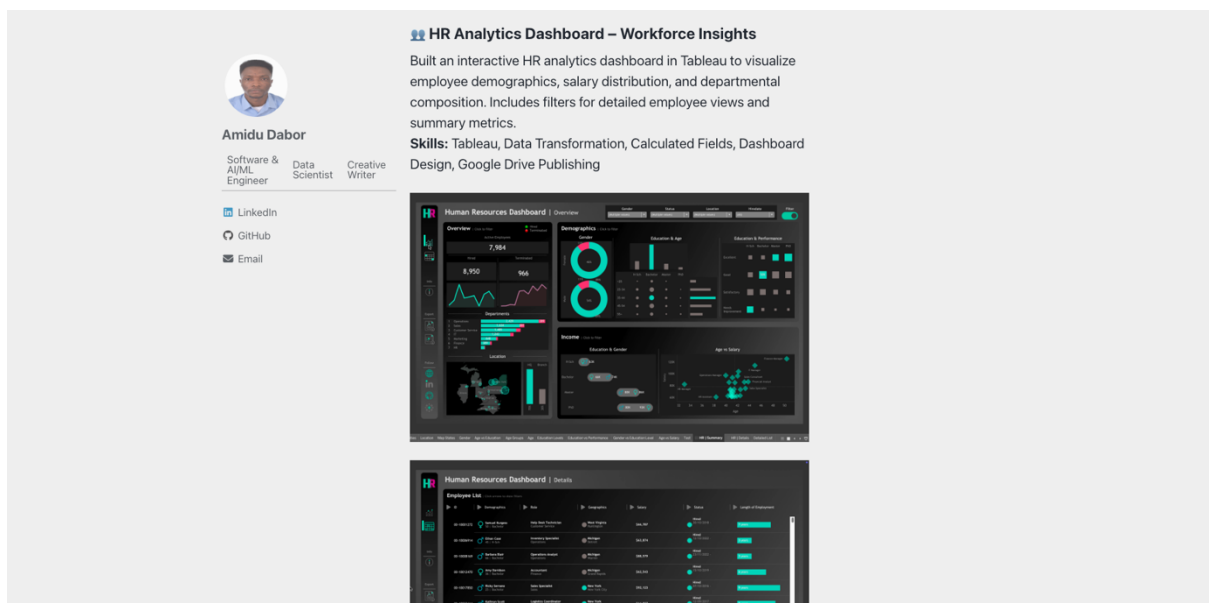
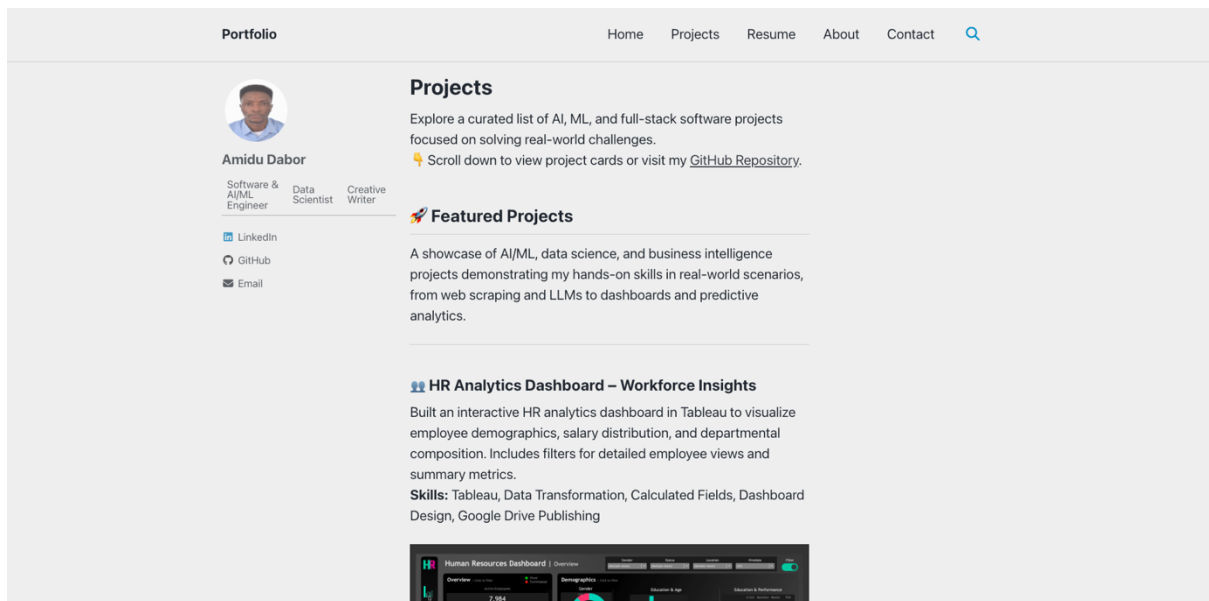
Steps Taken:

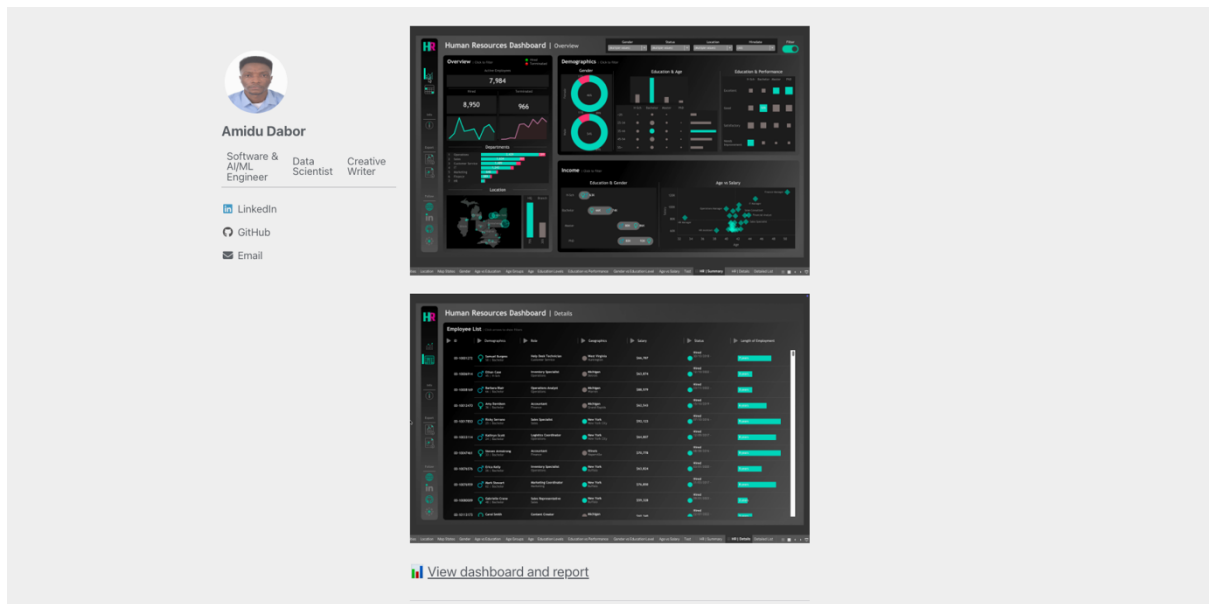
- Saved the final Tableau workbook as .twbx (packaged workbook)
- Uploaded the file to Google Drive
- Set sharing settings to "Anyone with the link"
- Confirmed public access by opening the link in an incognito browser

Dashboard Access Link:

https://drive.google.com/file/d/1Ln54U-aXyKIRw_iUpsP4iPTXVXYfWui-/view?usp=sharing

3.0 PORTFOLIO WEBSITE | HR ANALYTICS DASHBOARD - TABLEAU PROJECT





4.0 CONCLUSION

This Tableau project offered valuable hands-on experience in the complete data visualization process, from preparing and transforming raw HR data to designing interactive dashboards and publishing the final product. It strengthened my ability to translate data into meaningful stories that align with stakeholder goals. Through this project, I learned the importance of incorporating interactivity to enhance user engagement, using calculated fields to derive deeper insights, and applying effective design principles to make dashboards more intuitive and impactful. Publishing the final work as part of a public portfolio not only demonstrated technical proficiency but also added credibility to my data analytics skill set.

5.0 REFERENCES

1. Data with Baraa. (2025). [Tableau HR Dashboard Project](#)
2. Tableau Public Documentation. (2025). <https://help.tableau.com/>
3. YouTube. (2025). “Complete HR Tableau Project End-to-End” by Baraa